C4.2 What are the different types of polymers? *(separate science only)*

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*
Polymers are long chain molecules that occur naturally and can also be made synthetically. Monomers based on alkenes from crude oil can be used to make a wide range of addition polymers that are generally known as 'plastics'. Addition polymers form when the double bonds in small molecules open to join the monomers together into a long chain. **Condensation polymers were developed to make materials that are substitutes for natural fibres such as wool and silk.** **Condensation polymers usually form from two different monomer molecules which contain different functional groups. The OH group from a carboxylic acid monomer and an H atom from another monomer join together to form a water molecule. Monomers that react with carboxylic acid monomers include alcohols (to make polyesters) and amines (to make polyamides). To make a polymer, each monomer needs two functional groups. The structure of the repeating unit of a condensation polymer can be worked out from the formulae of its monomers and vice versa.**	1. recall the basic principles of addition polymerisation by reference to the functional group in the monomer and the repeating units in the polymer 2. deduce the structure of an addition polymer from a simple monomer with a double bond and vice versa **3. explain the basic principles of condensation polymerisation by reference to the functional groups of the monomers, the minimum number of functional groups within a monomer, the number of repeating units in the polymer, and simultaneous formation of a small molecule** ***ⓘ Learners are not expected to recall the formulae of dicarboxylic acid, diamine and diol monomers***

Linked learning opportunities

Specification links:

- The extraction and processing of crude oil, including the formation of alkenes by cracking. (C3.4)
- Functional groups including carboxylic acids and alcohols. (C3.4)

Practical work:

- Cracking poly(ethane) and testing the gas for unsaturation.
- Testing properties of different polymer fibres.

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Many natural polymers are essential to life. Genes are made of DNA, a polymer of four nucleotide monomers. Proteins (which are similar in structure to polyamides) are polymers of amino acids. Carbohydrates, including starch and cellulose, are polymers of sugars.	4. recall that DNA is a polymer made from four different monomers called nucleotides and that other important naturally-occurring polymers are based on sugars and amino-acids	**Specification links:** • Structure and function of DNA, and protein synthesis (B1.1) • The synthesis and breakdown of carbohydrates and proteins (B3.3) **Practical work** • Breaking down starch using an enzyme and using food tests to identify starch and sugars